

Submission — Full Stack Intern Technical Assessment

Project: Taskflow — Task & Team Management Platform
Candidate: Aniruddha Sutawane
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1. Links

Item	Link
GitHub Repository	https://github.com/aniruddha07S/task-management-platform
Live Frontend (Vercel)	https://task-management-platform-sepia.vercel.app
Live Backend (Render)	https://task-management-platform-wbvq.onrender.com
README (setup, API docs, screenshots)	README.md
API Collection (Postman v2.1)	Taskflow.postman_collection.json

Note: The backend is hosted on Render's free tier and sleeps when idle. The **first request may take up to ~50 seconds**; after that it responds normally.

2. Login Credentials (deployed website)

Role	Email	Password
User	testuser@example.com	Test@1234
Admin	admin@example.com	Admin@1234

Both accounts are pre-seeded in MongoDB Atlas and verified working on the live deployment. The login page also has "Use User" / "Use Admin" buttons that fill these in automatically.

3. Tech Stack

Layer	Technology
Frontend	React 19, Vite, React Router, Axios, Redux Toolkit, Tailwind CSS v4
Backend	Node.js, Express 5, JWT, bcrypt
Database	MongoDB Atlas (Mongoose)
Deployment	Vercel (frontend), Render (backend), MongoDB Atlas (database)

All three tiers are live and connected.

4. Requirements Checklist

Authentication

Requirement	Status	Implementation
Register	✓	<code>POST /api/auth/register</code> + Register page
Login	✓	<code>POST /api/auth/login</code> returns JWT
JWT authentication	✓	<code>server/middleware/auth.js</code> validates <code>Bearer</code> token
Protected routes	✓	Backend middleware + frontend <code>ProtectedRoute</code>
Logout	✓	Clears Redux state and stored token
Remember Me	✓	Checked → 30-day token in <code>localStorage</code> ; unchecked → 1-day token in <code>sessionStorage</code>
bcrypt password hashing	✓	<code>bcrypt.hash(password, 10)</code>

Dashboard

Requirement	Status	Implementation
Sidebar	✓	<code>Sidebar.jsx</code> — task lists with live counts, user, theme toggle, logout
Navbar	✓	Sticky toolbar with title, search and "New Task"
Cards: Total / Pending / Completed / In Progress	✓	<code>StatCard.jsx</code> , data from <code>GET /api/tasks/stats</code>

Task Module

Requirement	Status
Create	✓
Edit	✓
Delete (with confirmation dialog)	✓
View Details	✓ <code>TaskDetail.jsx</code>
Fields: title, description, priority, due date, status, assigned user	✓

Search & Filters

Requirement	Status
Search by title (debounced)	✓
Filter by status	✓
Filter by priority	✓
Sort by date	✓

REST APIs (all JSON, protected routes validate JWT)

Method	Endpoint	Purpose
POST	/api/auth/register	Create a new user account
POST	/api/auth/login	Authenticate and return a JWT
GET	/api/tasks	Retrieve tasks (search / filter / sort / paginate)
GET	/api/tasks/:id	Retrieve a single task
POST	/api/tasks	Create a task
PUT	/api/tasks/:id	Update a task
DELETE	/api/tasks/:id	Delete a task
GET	/api/tasks/stats	(extra) Counts per status
GET	/api/auth/users	(extra) Users for task assignment

Database

Requirement	Status
Users and Tasks stored in MongoDB Atlas	✓
Mongoose models with types and validation	✓ enums, required fields, unique email, email format
Relationship: task → assigned user	✓ assignedUser: ObjectId ref 'User' , populated in responses

Required React Concepts

Concept	Where used
<code>useState</code> , <code>useEffect</code>	Dashboard, TaskForm, Login, Register
<code>useMemo</code>	Dashboard (selected task), ThemeProvider (context value)
<code>useCallback</code>	All Dashboard handlers passed to memoized children
<code>React.memo</code>	TaskCard, StatCard, Sidebar, Segmented, Pagination
Custom Hooks	<code>useDebounce</code> , <code>useTheme</code>
Context / Redux	Redux Toolkit (<code>authSlice</code> , <code>tasksSlice</code>) + Context API (theme)
Lazy Loading + Suspense	Route-level <code>React.lazy</code> with a Suspense loader
Validation	Required fields, email format, password rules + confirm, duplicate users, invalid JWT → auto-logout

Error Handling

Code	Scenario handled
404	Unknown task ID, unknown route
400	Missing fields, invalid IDs, invalid filter values, duplicate email, wrong credentials
Network	Axios interceptor shows a clear "cannot reach the server" message
401	Missing/invalid/expired token → session cleared, redirect to login

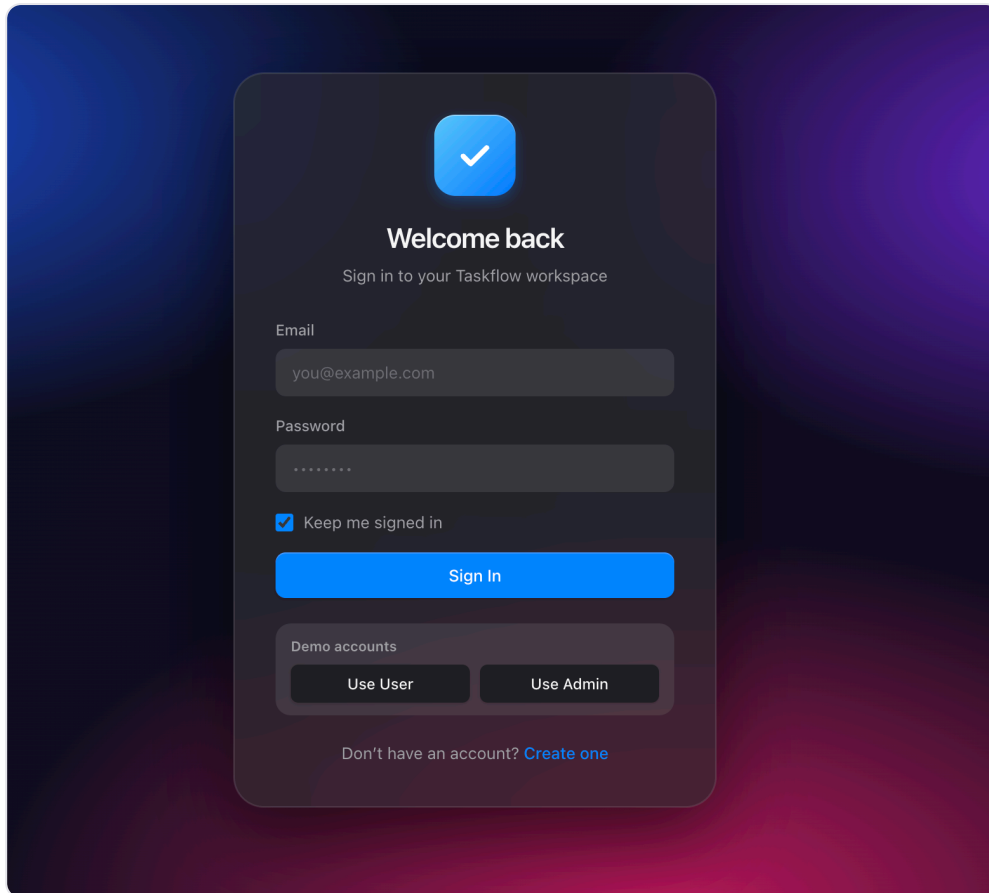
Folder Structure

Matches the expected structure: - `client/src/` → `components/` , `pages/` , `hooks/` , `services/` , `utils/` , `store/` - `server/` → `controllers/` , `models/` , `routes/` , `middleware/` , `config/`

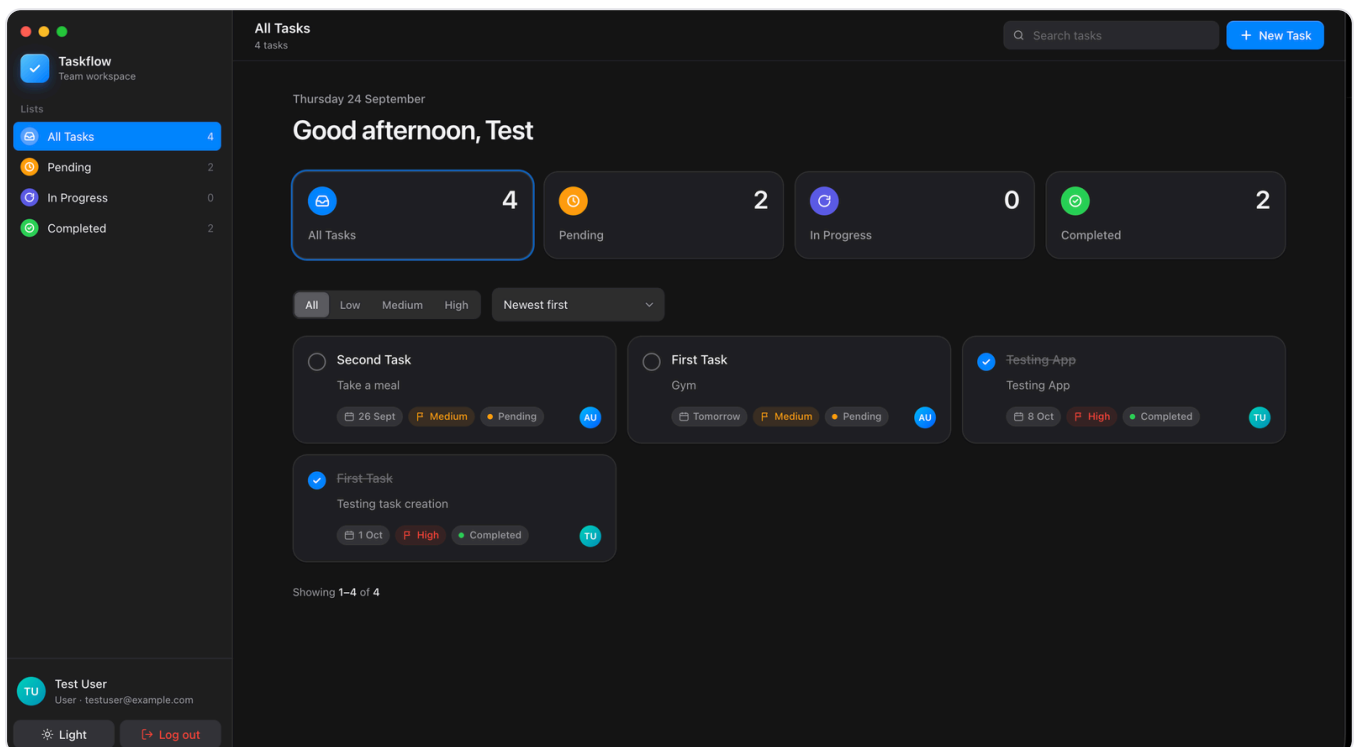
5. Bonus Features (4 implemented)

Feature	Details
Dark Mode	Toggle in sidebar and login page; persisted; follows system preference by default
Toast Notifications	Feedback on create, update, delete, complete and logout (<code>react-hot-toast</code>)
Pagination	Server-side <code>page / limit</code> with total count; stats endpoint keeps counts accurate across pages
Docker	<code>server/Dockerfile</code> , <code>client/Dockerfile</code> (multi-stage, nginx) and <code>docker-compose.yml</code>

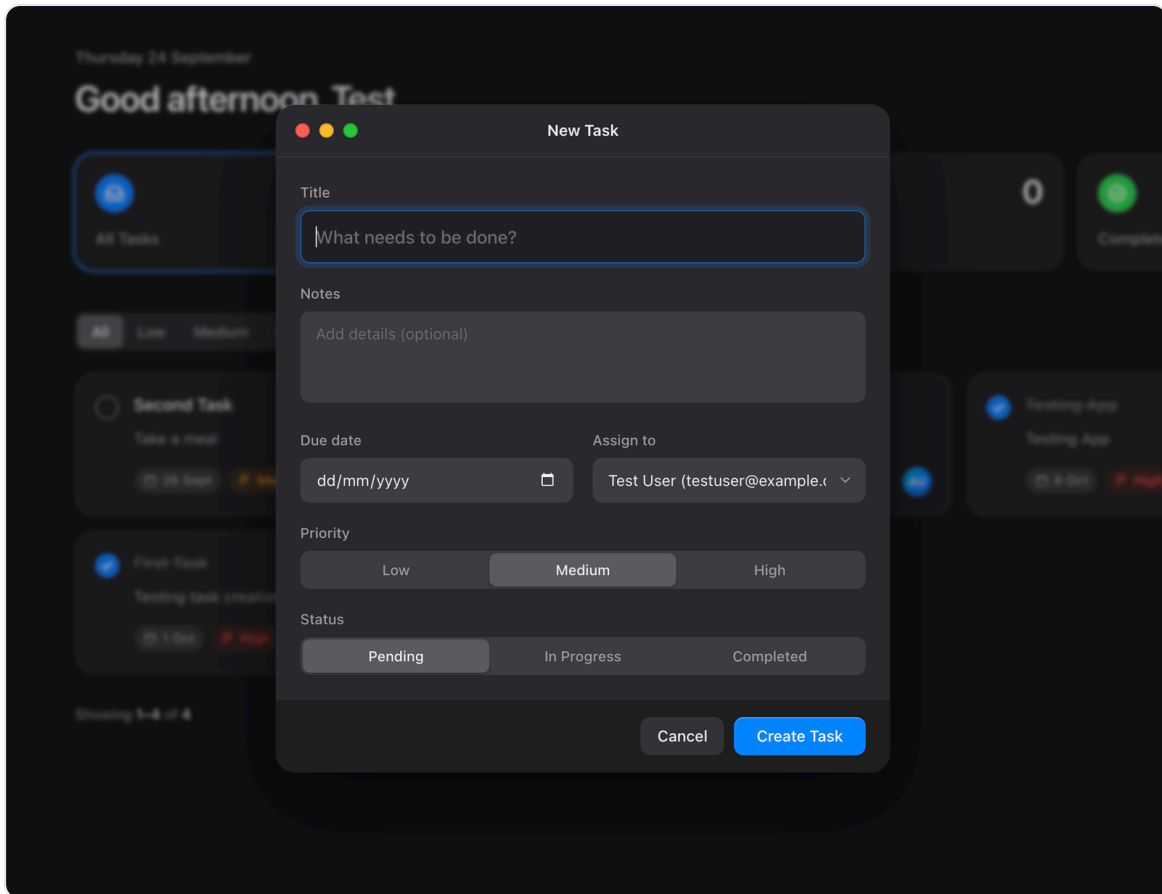
6. Screenshots



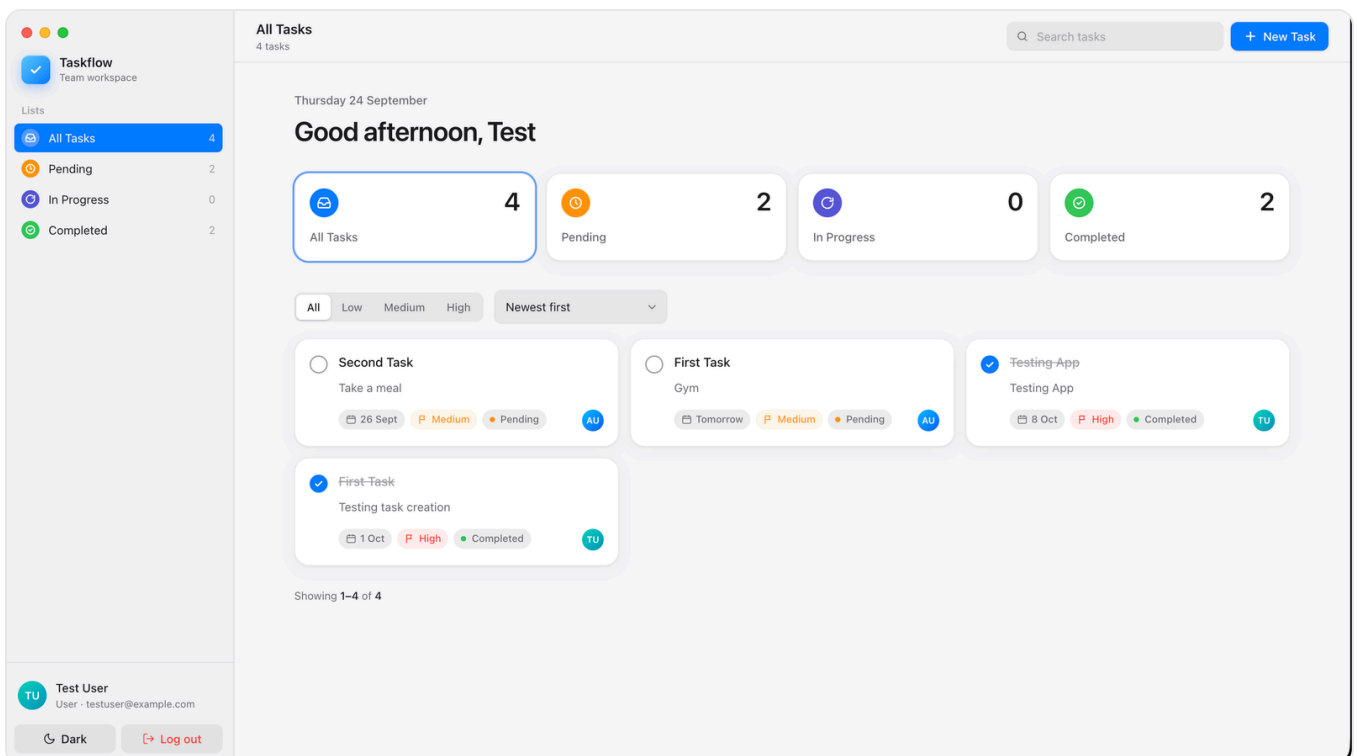
Login page (dark mode)



Dashboard — dark mode



Create task window



Dashboard — light mode

7. How to Evaluate Quickly

1. Open the **live frontend** and click **Use User** → **Sign In** (allow up to ~50s on the first request).
2. Create a task with **+ New Task**, click the card to view details, tick the circle to complete it, edit it, then delete it.
3. Try the search box, the sidebar lists (status), the priority filter and the date sort.
4. Toggle **Dark / Light** in the sidebar.
5. Import `Taskflow.postman_collection.json` into Postman and run the collection — **Login** stores the token automatically.

Local setup instructions are in the [README](#).